Q1.

(a) The average life span of a lion in a zoo is 22 years.
The average life span of a lion in the wild is 17 years.

Suggest **two** reasons why lions live longer in a zoo than in the wild.

1. .....

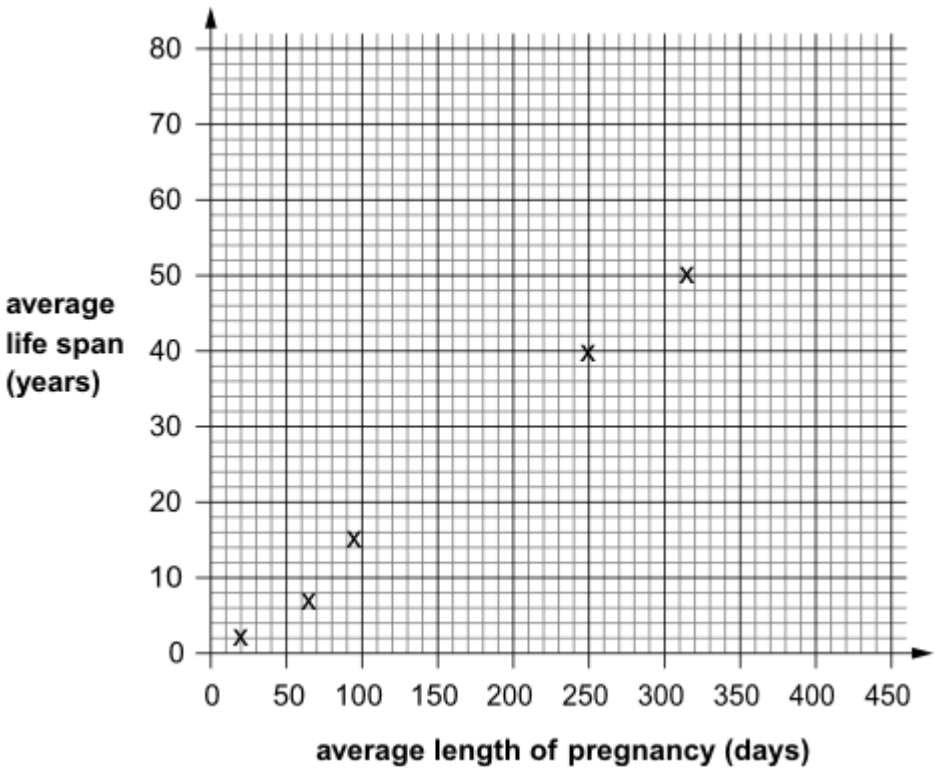
2. .....

1 mark

(b) John found the following data about five mammals.

mammal	average length of pregnancy (days)	average life span (years)
mouse	20	2
guinea pig	65	7
leopard	96	15
chimpanzee	250	40
whale	315	50

He plotted points using data from the table.



(i) Using the points John plotted, draw a line of best fit.

1 mark

(ii) From the graph, describe the relationship between the average length of pregnancy and the average life span.

.....

.....

1 mark

(c) John found data about three other mammals.

mammal	average length of pregnancy (days)	average life span (years)
Human	266	72
Horse	340	25
Giraffe	440	17

(i) Plot these **three** points on the graph above.

2 marks

(ii) Do these points fit the relationship you described in part (b) (ii)?
Tick the correct box.

yes ☐ no ☐

Use the graph to give a reason for your answer.

.....

.....

1 mark
maximum 6 marks

Q1.

(a) any **two** from

both *answers are required for the mark*

- less **or** no competition for food
- plenty of food **or** water **or** they get fed
  *accept 'good **or** balanced diet'*
  'they are looked after' is insufficient
- they have shelter
  accept 'protected from bad weather'
  'protection' is insufficient because it is not specific
- veterinary **or** medical treatment
  accept 'they are vaccinated'
  'they get fewer diseases' is insufficient
  *do **not** accept 'they are less likely to get a disease'*
  accept 'they are kept healthier'
  'they are healthier' is insufficient
- they are less likely to be injured **or** killed **or** there is no hunting of lions
  accept 'they do not fight as much'
  accept 'there are no predators of lions'

1 (L5)

(b) (i)
- an appropriate line of best fit
  *accept a curved **or** straight line*

1 (L6)

(ii) any **one** from

- animals with longer pregnancies live longer
- the shorter the pregnancy the shorter the life span
  accept 'the shorter the life span the shorter the pregnancy'
  a comparative answer is required

1 (L5)

(c) (i)
- all three points plotted correctly
  accept a tolerance of ± half a small square
  if all three points are correct, award two marks
  *if one **or** two points are correct, award one mark*

2 (L5)

(ii)
- no ✓
  if more than one box is ticked, award no mark
  both *the answer and the correct explanation are required for the mark*

any **one** from

- the data is more scattered
  accept 'they do not go up in a straight line'

- there is no link between the number of days pregnant and the expected life span
  accept 'they do not fit the line of best fit'

- the human has the longest life span but the shortest pregnancy

- the giraffe has the shortest life span but the longest pregnancy
  *accept 'giraffes **or** horses have a longer pregnancy than humans but a shorter life span'*

- the (new) points show the longer the pregnancy the shorter the life span
  accept the converse
  accept 'they are opposite to the other results'

1 (L6)

[6]